The Best Visualization Strategy for the Relationship Between X and Y

PSY 755 · Summer Project · Reseach memo

To: DSHBeauties group members **From:** Weizhu Pan **Date:** 27/7/2026

Re: *Which visual strategy best represents the potential relationship between the variable x and y ?*

This memo evaluates different visual strategies for illustrating the potential relationship between communication apprehension and public transportation usage. Given the complex nature of the dependent variable, which can be operationalized as continuous midpoint estimates, ordinal categories, or binary user status, we compare four distinct visualization approaches to assess their strengths and limitations for different data types and analytical contexts.

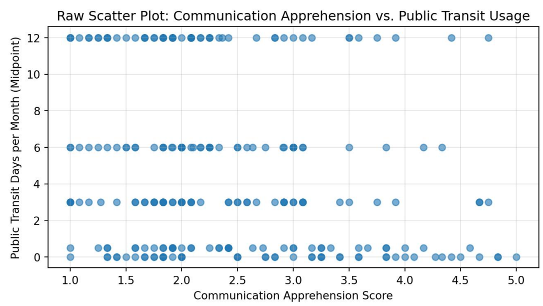

Figure 1

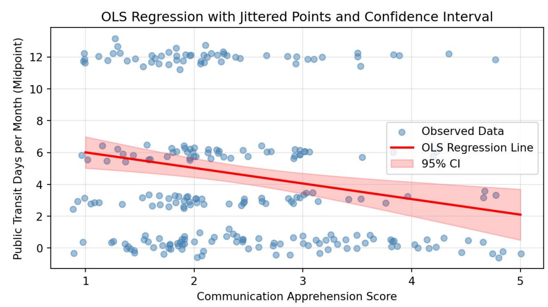

Figure 2

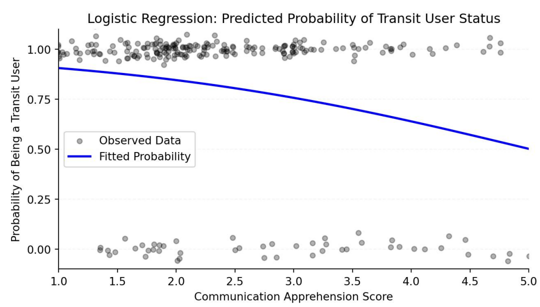

Figure 3

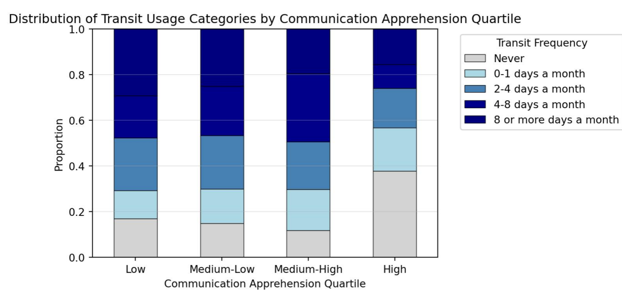

Figure 4

The four visualization strategies evaluated in this memo each serve different analytical purposes and are suited to different forms of the dependent variable:

- Raw scatter plots (Figure 1) are useful for initial data exploration but suffer from severe overplotting with discretized outcomes
- Jittered OLS regression (Figure 2) provides clear linear relationship information with confidence intervals but assumes unbounded continuous outcomes
- Logistic regression sigmoid curves (Figure 3) are theoretically appropriate for binary outcomes but discard frequency information
- Proportional stacked bar charts (Figure 4) preserve ordinal information and show distributional patterns but require discretization of continuous predictors

The choice of visualization should be guided by the specific research question, the nature of the forms of dependent variable, and the intended audience. For comprehensive analysis, researchers should consider presenting multiple complementary visualizations to fully capture the complexity of the relationship between communication apprehension and public transportation usage.